Transformations of Functions

Reference page | Original worksheet page 1

Complete worked solutions for Examples 1-9. Problem numbering follows the original worksheet. This is an independently prepared answer key, not an official publisher key.

The point-mapping rule

If (u, v) is on y = f(x), then for **y = a f(b(x - h)) + k**, with a and b nonzero, the corresponding point is:

(u, v) → (h + u/b, av + k)

Solve the inside expression for the new x-coordinate; apply the outside operations to the y-coordinate.

The four examples on the reference sheet

Function	Transformation	Point mapping
g(x) = f(x) - 2	Down 2 units	(u, v) → (u, v - 2)
g(x) = f(x + 2)	Left 2 units	(u, v) → (u - 2, v)
g(x) = -2f(x)	Vertical stretch by 2; reflect across the x-axis	(u, v) → (u, -2v)
g(x) = f(-x)	Reflect across the y-axis	(u, v) → (-u, v)

Domain, range and interval endpoints

Domain: transform the original x-values using the input mapping. **Range:** transform the original y-values using the output mapping.

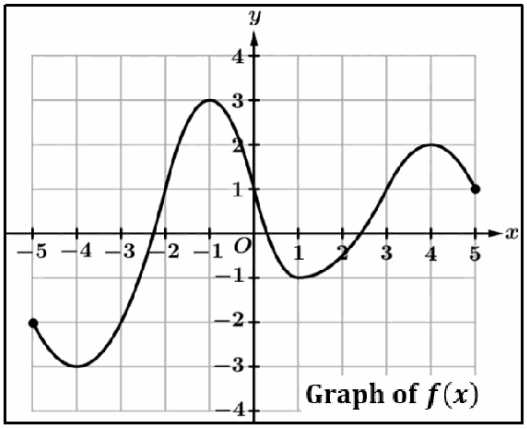
When multiplying by a negative number, reverse inequality signs. Preserve open and closed endpoints. A graph reflection may reverse the order of endpoints, so write the smaller endpoint first.

For graph-based increasing/decreasing intervals, this key uses open intervals between turning points. Example 9 retains the endpoint information explicitly supplied in the question.

Reading and moving the graph

Examples 1-2 | Original worksheet page 2

From the original graph:
Endpoints: (-5, -2), (5, 1)
Minimum value: -3
Maximum value: 3
f(-1) = 3; f(4) = 2



Example 1

Domain: [-5, 5]. Range: [-3, 3]. Both endpoints are included.

Example 2: g(x) = 2f(x - 1) + 3

Point mapping: **(u, v) → (u + 1, 2v + 3)**.

(a) Domain: [-4, 6]. Range: [-3, 9].
Shift [-5, 5] right by 1. For the range: 2[-3, 3] + 3 = [-3, 9].

(b) (4, 2) → (5, 7).
The new coordinates are 4 + 1 = 5 and 2(2) + 3 = 7.

(c) g(0) = 9.
g(0) = 2f(-1) + 3 = 2(3) + 3 = 9.

(d) Exactly one solution.
g(x) = 0 requires f(x - 1) = -3/2. The line y = -1.5 meets the original graph once, on its rising branch between x = -3 and x = -2. The left descending branch stays below -1.5; the later valley has minimum -1.

(e) Increasing on (-3, 0) and (2, 5).
f increases on (-4, -1) and (1, 4). Shift these intervals right by 1; the positive vertical factor preserves increasing behavior.

Reflection and table values

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Example 3: h(x) = -2f(x + 3) - 1

Point mapping: **(u, v) → (u - 3, -2v - 1)**. The source labels two different parts "a"; both are identified below.

(a, domain/range) Domain: [-8, 2]. Range: [-7, 5].
For the range, y = 3 becomes -7 and y = -3 becomes 5.

(a, concavity) Approximately (-5, -3) and (0, 2).
From the drawing, f is concave down approximately on (-2, 0) and (3, 5). The negative vertical factor reverses concavity, and the input shift subtracts 3 from each x-coordinate.

Graph-reading qualification: the inflection points are not labeled or specified by a formula. The boundaries near -2, 0 and 3 for f, and therefore near -5, -3 and 0 for h, are visual estimates.

(b) Increasing on (-8, -7), (-4, -2), and (1, 2).
f decreases on (-5, -4), (-1, 1), and (4, 5). Shift left 3 and reverse increasing/decreasing behavior.

(c) x = -7.
-2f(x + 3) - 1 = 5 gives f(x + 3) = -3. The graph attains -3 only at input -4, so x + 3 = -4.

Example 4: n(x) = -3m(2x) + 5

x	0	1	2	4	8
m(x)	**3**	3	**5/3**	5	-12
n(x)	-4	0	**-10**	**41**	13

Missing entries are bold.
-4 = -3m(0) + 5 → m(0) = 3.
0 = -3m(2) + 5 → m(2) = 5/3.
n(2) = -3m(4) + 5 = -15 + 5 = -10.
n(4) = -3m(8) + 5 = 36 + 5 = 41.

Do not pair the same x-column automatically: n(x) uses m(2x). For instance, n(8) = 13 implies m(16) = -8/3, outside the displayed m-table.

Polynomial and rational functions

Examples 5-7 | Original worksheet page 4

For Examples 5 and 6: **$f(x) = -2(x + 3)^2(x - 1)$**.

Example 5: g(x) = f(-x)

$g(x) = -2(3 - x)^2(-x - 1) = \mathbf{2(x - 3)^2(x + 1)}$.

(a) Zeros: x = -1 and x = 3. The zero at 3 has multiplicity 2.
(b) g(x) < 0 on (-∞, -1). Away from its zero, the squared factor is positive, so the sign is the sign of x + 1.

Example 6: h(x) = 2f(3x)

$h(x) = -4(3x + 3)^2(3x - 1) = \mathbf{-36(x + 1)^2(3x - 1)}$.

(a) h(1) = -288.
$h(1) = 2f(3) = 2[-2(6)^2(2)] = -288$.

(b) h(x) ≥ 0 on (-∞, 1/3].
For x < 1/3, the factor 3x - 1 is negative and the leading factor is negative, giving a nonnegative product. Include both zeros x = -1 and x = 1/3.

(c) y-intercept: (0, 36).
$h(0) = 2f(0) = 2[-2(3)^2(-1)] = 36$.

Example 7: k(x) = -4r(-x/2)

r(t) = (t + 1)/(t + 4), with **t ≠ 3, -4** retained from the original denominator. Therefore
k(x) = -4(x - 2)/(x - 8), with x ≠ -6, 8.

(a) Vertical asymptote: x = 8. Hole: x = -6.
Solve -x/2 = -4 and -x/2 = 3 respectively. The hole is at (-6, -16/7). Cancelling a factor does not restore its excluded input.

(b) k(x) ≥ 0 for x in [2, 8).
The sign of -4(x - 2)/(x - 8) is negative for x < 2, positive for 2 < x < 8, and negative for x > 8. Include the zero 2 and exclude the asymptote 8. The hole -6 is outside the solution interval.

Cubic data and interval mappings

Examples 8-9 | Original worksheet page 5

Example 8: m(x) = -3p(x + 4)

(a) Translate the graph of p **left 4** units, stretch vertically by a factor of **3**, and reflect across the **x-axis**. Point mapping: (u, v) → (u - 4, -3v).

(b) One zero is x = -2. Since p(2) = 0, set x + 4 = 2.

(c) Average rate of change: -6.
m(0) = -3p(4) = -9 and m(2) = -3p(6) = -21.
[m(2) - m(0)] / (2 - 0) = (-21 + 9) / 2 = -6.

(d) Domain: (-∞, ∞). Range: (-∞, ∞).
A nonconstant cubic polynomial has all real inputs and all real outputs. These nonzero transformations preserve that property.

(e) Directly from the table: x = -4.
m(x) = 3 means p(x + 4) = -1, and p(0) = -1.
Complete solution using the stated cubic condition:
x = -4, 5 - √73, or 5 + √73. See the derivation on page 6.

Example 9: q(x) = 4 - 2n(5 - x)

Set u = 5 - x, so **x = 5 - u**. The output mapping is **v → 4 - 2v**.

(a) Domain: [-4, 11]. Range: (-16, 20].
-6 ≤ 5 - x ≤ 9 gives -4 ≤ x ≤ 11.
For -8 ≤ v < 10, applying 4 - 2v gives -16 < q(x) ≤ 20.

(b) Increasing on (-4, 0] and [4, 11]; decreasing on [0, 4].
The input reflection and the negative output factor each reverse monotonicity, so their effects cancel. Map every interval using x = 5 - u:

Original n-interval	Corresponding q-interval
Increasing: [-6, 1]	Increasing: [4, 11]
Increasing: [5, 9)	Increasing: (-4, 0]
Decreasing: [1, 5]	Decreasing: [0, 4]

The source excludes u = 9 from its increasing interval, although n(9) exists. Thus x = -4 is excluded from that guaranteed increasing interval. Its endpoint behavior cannot be added without more information.

Why Example 8(e) has three solutions

Teacher note | Completing the cubic argument

The table provides five exact values of a function explicitly described as a cubic polynomial. Those values determine a unique cubic; they are not merely isolated facts about an otherwise unspecified function.

1. Recover the cubic

Let $p(t) = At^3 + Bt^2 + Ct + D$. The table gives $p(0) = -1$, so $D = -1$. Using $p(-2) = 1$ and $p(2) = 0$:

$-8A + 4B - 2C = 2$
$8A + 4B + 2C = 1$
Adding gives $8B = 3$, hence $B = 3/8$.

Using $p(4) = 3$ gives $64A + 16B + 4C = 4$.
Substitute $B = 3/8$ and combine with $8A + 2C = -1/2$:
$A = -1/48$ and $C = -1/6$.

$p(t) = -t^3/48 + 3t^2/8 - t/6 - 1$

The remaining table value is consistent: $p(6) = -216/48 + 108/8 - 1 - 1 = 7$. Any two cubics agreeing at four distinct inputs must be identical, because their difference would otherwise have more zeros than its degree.

2. Solve p(t) = -1

$-t^3/48 + 3t^2/8 - t/6 = 0$
Multiply by -48: $t^3 - 18t^2 + 8t = 0$.
Factor: $t(t^2 - 18t + 8) = 0$.

$t = 0, 9 - \sqrt{73}$, or $9 + \sqrt{73}$.

3. Apply the input shift

Since $t = x + 4$, subtract 4 from each input:

$x = -4, 5 - \sqrt{73}$, or $5 + \sqrt{73}$.

Approximately: $x = -4, -3.5440$, or 13.5440.

Teaching use: $x = -4$ is the answer immediately visible from the table and is sufficient if the prompt is interpreted as “give one x-value.” If “for which x-value” is intended to request all solutions, the stated cubic condition gives the three values above.

Source: APPC 1.12 Notes - Transformations of Functions, created by Bryan Passwater, supplied worksheet, pages 1-5. The graph on answer-key page 2 is reproduced from the supplied worksheet for reference.